Fire and Flames

Event platform and operations plan

College review edition • 27 October 2026

One platform for E-Fest by ED Cell and Kurukshetra esports by Srustiprosar, with secure registration, verified payments and accountable event-day operations.

Prepared for Srusti Academy of Management & Technology (Autonomous)

Documented E-Fest dates: 13–14 November 2026. Participation is open to students from other colleges.

This review edition contains proposed workflows and acceptance criteria. Official payment details, esports rules and several college policies remain approval items. It does not represent a deployed or tested application.

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Companion deliverable: Fire and Flames Design Boards. Editable sources, structured catalogue, acceptance register and asset provenance are included in the planning folder.

Product and scope

Build one trustworthy event platform with two distinct public experiences and one controlled operations system. Students should understand the organizer, event requirements, price and payment status before registering. Staff should see only the records and actions their assigned duties require.

Confirmed foundation

Fire & Flames is the E-Fest of Srusti Academy of Management & Technology (Autonomous), organized by ED Cell on 13–14 November 2026. The shared entrance also presents Kurukshetra esports, Free Fire and BGMI, organized by Srustiprosar per the user's instruction. Dates for individual esports sessions require confirmation. Students from other colleges are welcome. Teachers have a separate sign-in entry but do not automatically receive management permissions.

Core requirements

ID	Requirement	Release evidence
R01	Two visible organizer-labelled hero panels and deep-linked event pages	Desktop and mobile navigation reaches the correct event
R02	Verified student identity, college profile and self-service registration	External-college student completes the full flow
R03	Teams, invitations, roster consent and membership limits	Duplicate membership and invalid sizes blocked server-side
R04	Official college QR payment and controlled reconciliation	Payment evidence alone cannot produce a valid pass
R05	Teacher, volunteer, judge, admin and super-admin access	Denied action tests across roles and organizer scopes
R06	Volunteer QR attendance and duty assignment	Concurrent duplicate scans record one valid attendance
R07	Event management, capacities, schedules, scoring and results	Published results trace back to locked scores and approval
R08	In-app and email transaction notifications	Outage, retry and deduplication tested
R09	Responsive, accessible, fast public and staff screens	Real phone, keyboard and reduced-motion checks
R10	Audit history, backups, restoration and incident controls	Restore rehearsal and attributable activity logs

Release scope

The first production release includes both landing paths, the six documented E-Fest activities, esports information pages, student/teacher authentication, team and solo intake, manual official-QR payment reconciliation, passes, online scanning, scoped operations, judging, controlled results, notifications and support. Paid esports intake opens only after its separate configuration is approved. This is a feature gate, not a fake free registration option.

Each event has its own registration. A student can enter multiple compatible activities; there is no implied all-event pass or combined fee. The initial checkout handles one event registration per order. Team price is charged once per team,

never once per member. Cross-organizer consolidated checkout is deferred to avoid ambiguous payment allocation.

Proposed technology direction

Prefer a separate React and TypeScript application, a managed PostgreSQL backend, managed authentication, private file storage and server-side operations. Supabase and Vercel are the proposed fit based on Crossfire's existing approach; versions, account ownership, hosting plan and costs must be assessed at implementation. Do not reuse Crossfire's live database or keys.

Sites was assessed as an alternative. Its documented hosted identity is ChatGPT sign-in. It is not the selected production path for the requested college email and teacher/student flow unless the college explicitly accepts that identity model. No Sites project is registered during planning.

Success measures and limits

Acceptance targets: no unauthorized cross-track access; no duplicate payment allocation; no confirmed entry without verified funding and valid roster; no duplicate same-session attendance; all operational changes auditable. Target a typical verified scan response below two seconds on a working campus connection. Provisional load test: 500 concurrent browsing sessions, 50 concurrent registration mutations and 20 scanners; validate actual expected attendance before sizing hosting.

Public loading should target LCP at or below 2.5 seconds, INP at or below 200 milliseconds and CLS at or below 0.1, measured at the 75th percentile where field data is available. These are targets, not measurements of an app that already exists. Reference: https://web.dev/articles/defining-core-web-vitals-thresholds.

Explicit exclusions

No automated bank verification without a supported verified integration; no live esports game telemetry or anti-cheat claim; no guaranteed notification delivery; no automatic Instagram vote scraping; no offline scan claiming final success. Meals, certificates and guest hospitality are optional modules, not promises inherited from Crossfire. The college must enable and define them before they appear publicly.

Event catalogue and source rules

Use the supplied rules document for detailed judging weights and the PDF for visual direction. Conflicts remain visible until an organizer signs off. Monetary values below are INR. Store integer paise in the future database.

E Fest registration and prizes

Activity	Team or capacity	Registration	Awards	Pool
IPL Auction	2–4 members; 8 teams in a pool	₹99 per team OR ₹59 per person	₹4,000 / ₹2,500 / ₹1,500	₹8,000
e-Trivia	2–4 members	₹99 per team OR ₹59 per person	₹2,000 / ₹1,500 / ₹1,000	₹4,500
Treasure Hunt	2–4 members	₹99 per team OR ₹59 per person	₹3,000 / ₹2,500; discipline ₹1,500	₹7,000
Brand War	2–3 members	₹99 per team OR ₹59 per person	₹2,000 / ₹1,500 / ₹1,000	₹4,500
Srusti Bazar	10 stalls; member limit unspecified	₹299 per stall	Three awards of ₹1,000	₹3,000
Srusti Exhibition	10 entries; member limit unspecified	₹199 per entry or team	Three awards of ₹3,000	₹9,000

Total for these six activities is ₹36,000. Eight teams is a pool size for IPL, not necessarily the total event capacity. Do not enforce an invented capacity for trivia, treasure hunt or Brand War. Ten stalls and ten exhibition entries are the published capacities; overrides require a revised published configuration.

Esports

The PDF advertises Kurukshetra, Free Fire and BGMI with a combined esports prize pool of ₹20,000. Organizer attribution is Srustiprosar, supplied by the user. Entry fee, player count, substitutes, match format, server, device restrictions, scoring, tie-breaks, prize split and reporting times are unknown. Save unknown fees as null, never zero. Each game has a draft event record with registration disabled. Display “Registration details coming soon” until approved. Do not distribute ₹20,000 equally or advertise ₹56,000 as a confirmed festival-wide total without approval.

Required event rules

All participants register and pay before the activity; roster changes need organizer approval. Schedules, time limits, conduct and fair play apply. Disruption, cheating, disrespect or property damage can disqualify. Evaluator and committee decisions are final under the source rules. Prizes require completed activity and verified results.

IPL Auction uses an external evaluator and ranking by points. Budget, player pool, squad size, role constraints and scoring formula are not specified; publish a versioned rules supplement before play. Recognized bids are binding, overspending is prohibited and teams explain squad choices.

e-Trivia uses an internal evaluator and one point per correct entrepreneur answer. Quizmaster defines round formats, answer submission and time limits. Phones, internet and outside help are prohibited unless a round allows them. Publish whether response time affects score or only ties before the quiz.

Treasure Hunt rewards the first and second teams to solve all clues/checkpoints correctly in the least time. The discipline award is separate and discretionary, although shown in the third-prize column. Keep teams together unless a checkpoint allows separation; prohibit unsafe areas, disruption, tampering and unauthorized help. Server timestamps

plus marshal records determine ordered progress; GPS is not required or proof of completion.

Brand War uses an external evaluator. Teams complete a brand task/pitch within time, use original or credited content and answer questions respectfully. Time deductions must be defined before judging and recorded, not silently subtracted.

Bazar awards Best Stall by Jury, Most Popular Stall and Innovative Stall. The official ED Cell Instagram live poll determines Most Popular; organizers announce the voting window and verify results. The platform records verified poll evidence and approval, rather than replacing it with a website poll. Stalls manage setup, accurate pricing, cleanliness, safety, equipment and clear walkways.

Exhibition entries explain the problem, solution, users, value and development stage. External jury can request demos and ask about feasibility, costs and differentiation. Teams disclose third-party work and special power/setup needs. Multiple awards for one entry are not assumed.

Judging weights from the Word document

Rubric	Criterion and maximum marks	Total
Treasure discipline	Respect 25; instructions 25; fair play 25; coordination 25	100
Brand War	Brand accuracy 25; creativity 25; marketing/customer insight 20; pitch 20; teamwork/time 10	100
Bazar Best Stall	Quality/relevance 25; presentation 20; customer interaction 20; business/pricing 15; branding 10; preparedness 10	100
Bazar Innovative Stall	Novelty 30; usefulness 25; demonstration 20; sustainability where relevant 15; scalability 10	100
Exhibition	Problem clarity 15; originality 20; feasibility 20; prototype where applicable 20; user/market value 15; presentation/questions 10	100

Judge scores must fall within each criterion's maximum. The organizer must decide how “where applicable” is scored before publishing: no silent zero, silent omission or undocumented renormalization. Multiple-judge aggregation and tie-break policy also need approval; store both raw marks and the approved calculation version.

Conflicts to resolve

Exhibition summary names Best Startup / Business Concept, Best Working Prototype and Most Promising Idea. Both files' award tables instead name Best Problem identified, Best Working Prototype and Best Scalability idea. Propose the clearer summary names for review, but do not silently replace the source labels.

The PDF omits several weighted rubrics present in the DOCX, repeats the institutional description under About the Event, and contains spelling/formatting errors. Use accessible HTML text for corrected approved copy; preserve original source files for traceability. No official payment QR was supplied in these files.

Roles and access

Identity, affiliation and operational authority are separate. A person may be a student volunteer or a teacher judge, but staff permissions exist only through approved assignments. Selecting “Teacher sign in” is not a grant of admin rights.

Role definitions

Role	Access scope	Allowed duties	Never implied
Visitor	Public	View published events, rules, schedules and approved results	Participant directory or payment data
Student	Own profile and accepted memberships	Register, invite members as captain, submit own payment, view own passes	View other students or approve funds
Teacher	Own verified profile	View assigned coordination tasks; judge if separately assigned	Automatic admin or finance access
Volunteer	Active shift, event and station	Scan, limited lookup, attendance and incidents	Fees, funds approval, exports or scores
Judge	Assigned event and candidates	Draft and lock own rubric scores	Other judges' draft scores or payment evidence
Event admin	Assigned organizer and events	Registrations, teams, timetable, volunteer/judge duties, result proposals	Cross-organizer access or super-admin grants
Finance reviewer	Explicit additional permission and organizer scope	Reconcile deposits, reject evidence, propose refunds	Change beneficiary or own-payment approval
Super admin	Institution-wide explicit membership	Assign admins, approve settings, payment profile, result publication and incident recovery	Bypass audit or unilateral self-approval

Permission matrix

Action	Student	Teacher	Volunteer	Judge	Event admin	Super admin
Own profile and registration	Own	Own where eligible	Own student role	Own where eligible	Own where eligible	Own where eligible
View event participant roster	No	Assigned only	Minimal on-duty	Assigned candidates	Scoped	Institution
Check in participant	No	With duty	Assigned station	No	Scoped override with reason	Logged override
Verify payment	No	Finance grant only	No	No	Finance grant only	Finance grant; no self-review
Edit event setup	No	No	No	No	Draft scoped	Approve/publish
Assign volunteer or judge	No	No	No	No	Scoped	Institution
Grant admin or finance role	No	No	No	No	No	Yes with reason and MFA

Action	Student	Teacher	Volunteer	Judge	Event admin	Super admin
Submit scores	No	Judge grant	No	Assigned	Only if judge grant	Only if judge grant
Publish or correct results	No	No	No	No	Propose	Approve
Export personal data	Own	No	No	No	Separate export grant	Logged and p urpose-limited
Change payment beneficiary	No	No	No	No	No	Propose; second approver required

“Assigned” always means explicit current database assignment, not a frontend route or email suffix. Scope checks apply to organizer, event, duty window and record ownership on every request. A global role does not remove conflict-of-interest restrictions.

Account lifecycle

Students verify an email address, complete name, mobile, college, course/year and college-ID number only where needed for eligibility, and accept terms. No bulk public student list. Captains invite members with expiring single-use links; each member authenticates and consents. A captain cannot set another person's password or silently consent for them.

Teachers are invited or submit an access request for manual college verification. Staff invitation is issued to a specific address and role; first sign-in requires account setup and MFA enrollment. No shared volunteer accounts. Admins may assign already verified volunteers/judges only within their own events. Only super admins grant admin/finance privileges. Assignment expiry automatically removes effective permissions.

Provision the first super admin through a documented owner-controlled bootstrap outside public registration. Require a second named super admin for beneficiary changes and recovery. The last active super admin cannot be demoted without an approved replacement. There is no public role dropdown that grants elevated access.

Session and audit behavior

Staff MFA is enforced on the server and database, not just shown as a setup prompt. Proposed staff idle timeout: 30 minutes with reauthentication; require recent MFA for privilege grants, exports, refunds and beneficiary changes. Revoked roles must stop working immediately by checking active membership on each sensitive action; do not wait for a cached role claim to expire. Student sessions use normal managed-auth limits and explicit sign-out on shared devices.

Audit entries contain actor, effective role, scope, action, record, time, reason, before/after summary and request correlation ID. Exclude raw QR tokens, passwords, session tokens and full payment screenshots. Audit access is restricted and retention follows college policy. Application users cannot update or delete audit records.

Registration payments and QR workflows

Keep registration, payment, eligibility and attendance as distinct records. The UI must describe the real state and required next action. “Evidence submitted” does not mean “Paid”, and “Paid” does not automatically mean “Eligible to participate”.

Student and team journey

Browse either organizer track, select an event, read price/size/rules, sign in, complete college profile, choose team or individual mode where published, obtain member consent, review the roster and fee, accept the event rules version, reserve a slot, pay the displayed official beneficiary, submit a transaction reference and optional supporting receipt, await verification, and receive a confirmed event registration and individual pass.

The ₹99 team option covers one valid team. ₹59 individual intake is a separate product, not ₹59 multiplied automatically into a team fee. Proposed individual policy: coordinator pools individually paid applicants into valid teams with consent, with no extra ₹99 charge; no pass until allocation completes. This policy needs college approval before individual checkout opens. Until then show the documented individual price with “Individual registration policy awaiting confirmation”; do not accept unallocatable payments.

For team mode, enforce 2–4 members or 2–3 for Brand War before payment. Pending invitations must be resolved first. Once evidence is submitted, freeze the price and roster snapshot. A post-payment roster change requires admin review, member consent, eligibility recheck, audit and pass reissue for affected members. Moving captainship must not create a second order or payment.

Registration states

State	Entry condition	Next allowed actions
draft	Student starts own form	Edit, invite, abandon
awaiting_members	Valid captain, unresolved invitations	Accept/decline, revise roster
awaiting_allocation	Individual policy approved; allocation incomplete	Coordinator proposes team, students accept
payment_due	Eligible roster or approved solo-intake mode; slot held	Pay, submit evidence, cancel, expire
payment_review	Evidence submitted within hold	Reviewer verifies, requests correction or rejects
action_required	Evidence mismatch or incomplete information	Submit corrected evidence before deadline
confirmed	Verified full payment plus valid eligibility and allocated roster	Issue pass, attend, request change/cancellation
waitlisted	Capacity unavailable before collection	Accept a time-limited offer; no payment yet
expired	Reservation deadline elapsed without evidence	Recheck availability and start a new reservation
cancelled	Approved withdrawal or event cancellation	Refund workflow if required
disqualified	Authorized conduct/rules decision	Revoke event pass, record reason and appeal channel

Proposed hold: 30 minutes to submit evidence. Evidence submitted on time protects the slot through manual review. Escalate review older than 24 hours to finance; never silently expire a paid-looking submission. Capacity counts confirmed entries and active holds/reviews; waitlist and abandoned drafts do not count. Exact timer and support coverage need approval. Allocate the last slot transactionally so two users cannot buy it.

If a transfer arrives after a slot expires or an event fills, mark “Verified payment awaiting resolution”; do not force-confirm above capacity. Finance offers an approved reallocation or refund. This exception blocks automatic pass issuance and appears in a dedicated queue.

Payment configuration

The college supplies its official QR image, beneficiary name, UPI identifier and authorized receiving account confirmation through a controlled channel. Two named approvers verify decoded destination against college records and conduct an approved test transfer before activation. Store a versioned payment profile with image checksum, verified destination, effective time and approvals. Never recreate an official QR using guessed information.

Desktop checkout shows the verified QR, beneficiary, exact amount, order reference and a copyable UPI ID. Mobile also offers an approved UPI intent link if the college provides/verifies the required details, plus QR download/manual payment guidance. Static QR often does not encode the order amount, so show the exact amount prominently. Do not assume a UPI app return, intent callback, screenshot or typed UTR proves payment.

Payment credentials are separate from entry-pass QR tokens. Never place the college payment QR on an attendance pass. Never style a decorative QR as a functional one. The design boards use labelled boxes for missing QR assets.

Reconciliation state machine

Payment progresses from unpaid to evidence_submitted, then under_review, then verified, rejected or needs_information. Corrections append a new evidence version. Match transaction reference, credited amount, beneficiary, transaction time and settlement evidence from authorized bank/merchant records. Reviewer records source/reference and reason. No screen scraping of personal banking accounts is part of this plan.

Use a unique settled bank transaction identifier and receiving account combination to prevent the same credit funding two orders. Typed UTR uniqueness alone is not proof. Handle duplicated submissions idempotently. Reject a reviewer verifying their own order or a team they belong to; route to another reviewer. UI displays the verification history without exposing bank statements to students.

Underpayment, overpayment, missing reference, duplicate reference, wrong beneficiary, unreadable evidence and late payment each enter explicit exception queues. Do not silently change an order price to match a transfer. One verified transaction funds one order in the first release; split/combined bank transfers require finance resolution. Refund states are requested, approved, processing, completed or failed, with amount, reason and actual bank reference; clicking approve is not proof money was returned. Refund policy and service targets require college approval.

Pass issuance and scanning

A valid confirmed registration issues one pass per authenticated participant per event; the team captain also sees a roster summary. Each pass has a random opaque token of at least 128 bits, its hash stored server-side, expiry, version, event and participant association. The QR contains no name, phone, college ID, payment data or role. Display event, participant display name, team, date/time when published and a human-readable support reference beside the QR.

Volunteer signs in with MFA, starts an assigned shift, selects the authorized station/session, permits camera access and scans. Server checks token validity, current registration, revocation, event/session, duty assignment and attendance uniqueness in one transaction. Successful attendance returns a concise green confirmation with matching identity and time; repeated scan returns amber “Already checked in” with original time; wrong event, invalid or revoked pass returns a clear denial. Never reveal a full profile for an invalid token.

Uniqueness key is participant plus event session plus attendance action. Day-one check-in must not block a separately authorized day-two session; event entry is distinct from campus entry. Check-out/re-entry, if enabled, requires an explicitly modeled policy. Shared team passes must not mark absent teammates present. Copied valid QR images are mitigated by server replay protection and name/college-ID visual matching where required; opaque tokens alone do not prevent sharing.

If the camera fails, use a rate-limited support-reference lookup at the assigned desk and verify identity before marking attendance. Log manual entries and reason. On network loss show “Not recorded”; pause definitive check-in. Authorized fallback is a numbered paper exception ledger, later reconciled by a supervisor. Do not store a roster or unrestricted payment data on volunteer phones.

Treasure checkpoints and optional entitlements

Checkpoint completion is separate from event attendance. Assigned marshals validate a team's pass and checkpoint in order, recording server time and any approved penalty. The last checkpoint does not automatically publish a winner; the evaluator reviews correct completion and penalties. Optional meal/certificate records use their own eligibility and uniqueness rules and remain off until college approval.

Experience and brand specification

The public site should feel like the brochure brought into a clear, fast web experience: charcoal, metallic gold, ember orange, deep red and warm white. Management screens should emphasize readable data and unambiguous status. The attached design boards are static concepts, not working screens.

Two event entrances

Desktop hero shows two equal panels simultaneously, with a shared institution header and the event date above them. Left: “Fire & Flames / The E-Fest / Organized by ED Cell”, six activities, documented ₹36,000 activity pool, “Explore E-Fest”. Right: “Kurukshetra / Esports / Organized by Srustiprosar”, Free Fire and BGMI, advertised ₹20,000 esports pool, “Explore Esports”. Unknown fees show “To be announced”. Do not put a single ambiguous Register button over both panels or hide one track in an auto-rotating carousel.

At 360–430 px, use stacked compact hero panels in a clear order, each with its organizer label and action. Both are reachable immediately without dismissing an intro. A compact top-level track selector links to each section. Use separate URLs /e-fest and /esports; individual event pages live below the correct track. Selection persists through sign-in using a validated internal return path. Site-wide account view groups registrations by organizer.

Avoid compulsory loading animations, scroll hijacking, autoplay sound and heavy WebGL. Optional brief ember animation starts after content is usable, stops offscreen/in background and honors reduced motion. Deep links, back navigation and recovery after refresh must work without replaying an intro.

Design tokens

Token	Proposed value	Usage
Canvas	#0B0B0D	Public background
Surface	#161619	Cards and panels
Raised surface	#222226	Dialog and form panels
Primary text	#F7F2E8	Main copy on dark
Secondary text	#C4BFB5	Supporting labels
Ember	#FF781F	Primary action with near-black text
Gold	#F5BF42	E-Fest accent and highlights
Deep red	#9B1C17	Decorative brand accent, not small text
Border	#3A3530	Field boundaries
Success	#91E6B5	With “Verified” or “Recorded” text
Warning	#FFD57A	With reason and next action
Error	#FFB4AB	With message, not color alone

These are proposed web tokens matched to the PDF visually, not a claim that the brochure specifies exact hex codes. Use the extracted official logos without redrawing or recoloring. The ED Cell badge is only 160×160 in the source; render it modestly and request a higher-resolution official master for large print. Preserve source aspect ratios and transparency.

Use a condensed bold display typeface for short event headings and a readable sans-serif for content, forms and tables. Prototype boards use available Arial; final font family and self-hosting license must be verified. Minimum body size 16 px on phones; avoid compressed paragraph fonts from the brochure. Render fees, dates and requirements as HTML, not flattened images.

Layout and accessibility

Use an 8 px spacing base, 16–24 px phone gutters, 24–32 px desktop card gaps and a 1200–1280 px content width. Target 44×44 px controls or larger; mobile scan action 56 px. Keep visible keyboard focus, skip link, semantic labels, error summaries, descriptive button text and sufficient contrast. Gold/orange buttons use near-black text; test contrast, including disabled states. Tables convert to labelled record cards on narrow screens, with no clipped actions. Charts include text summaries.

Meet WCAG 2.2 AA as the acceptance objective, with manual keyboard/screen-reader review. Reference: https://www.w3.org/TR/WCAG22/. Practical target-size reference: https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum. Use reduced-motion preferences and never use animation to communicate the only copy of a status.

Screen inventory

Area	Required screens	Critical states
Public	Dual hero, track pages, event detail, rules, schedule, help, approved results	Loading, unavailable, closed, sold out, fee unknown
Authentication	Student sign-in, teacher sign-in, verification, reset, MFA, access request	Expired link, invalid credentials, disabled account, unassigned teacher
Student	Profile, college details, event choice, roster invitations, review, checkout, payment status, pass, history	Incomplete team, waitlist, correction needed, refunded, revoked pass
Volunteer	Duty selection, scanner, scan result, lookup, incident, shift summary	Wrong station, off duty, duplicate, denied camera, offline
Judge	Assigned candidates, rubric, save draft, submit lock, conflict declaration	Missing criterion, score limit, locked score, reopened revision
Admin	Overview, scoped event setup, roster, reconciliation, assignments, schedules, result proposals, broadcasts	Pending approval, conflict, partial failure, stale edit
Super admin	Institution overview, roles, settings approvals, beneficiary versions, exports, audits, recovery	Last-admin protection, dual approval, suspended track

Form and status design

Registration is a short stepper: event → profile → team → review → payment → status. Save drafts server-side for signed-in users. Keep one primary action per step and show the selected organizer/event/fee summary throughout. Preserve valid input after errors. Required fields are visibly labelled. Show exact fees rather than “starting from” during checkout.

Student dashboard shows “Action needed”, “Payment under review”, “Confirmed” and “Waitlisted” separately. A pass button is disabled with an explanation until all conditions are met. Teacher workspace shows assignments and an access-request route if none exist. Admin dashboard starts with review queues and operational exceptions, not vanity metrics. Super-admin dashboard emphasizes unresolved approvals and access risk.

Asset and performance budget

Use named source artwork from assets/brand, optimized derivatives from assets/web, and the generated background only for decorative atmosphere. Preload at most the primary hero background; lazy-load offscreen event cards. Suggested first-viewport image budget: 350 KB mobile and 700 KB desktop; initial route JavaScript target 200 KB gzip excluding deferred scanner/management modules. Set image dimensions to prevent shifts. Preserve a solid-color background when images fail. Do not send the 42 MB brochure as a page background; brochure download is explicit.

Copy examples

Primary actions: “Explore E-Fest”, “Explore Esports”, “Review registration”, “Submit payment details”, “View my pass”, “Start duty”. Payment message: “Your payment details are under review. We will notify you after verification.” Offline scanner: “Connection lost. Attendance has not been recorded.” Staff rejection: “This action is outside your assigned duty. Contact the event desk.”

Architecture and data contracts

Proposed design: a modular application with a public frontend, authenticated workspaces, server-side commands, transactional PostgreSQL, managed authentication, private object storage and an outbox worker. This is a specification; no service account or database has been provisioned.

Components and boundaries

Public published event content can be cached. Registration, funds, passes, scores, roles and private documents must remain authoritative on the server. Authentication establishes identity; database permissions and server-side ownership/scope checks authorize actions. Browser state is for UI preferences, never the source of payment or attendance truth.

Use a fresh Supabase project for Auth, PostgreSQL and private Storage, and a separately owned web hosting project such as Vercel, subject to college approval. Sensitive workflows run through narrow server endpoints or reviewed database procedures. Transactional email comes from a college-approved provider and sender domain. SMS/WhatsApp and bank APIs are optional later integrations, not assumed available. Assess plan limits, database recovery, email quotas and secret ownership before committing to vendors.

OWASP authorization guidance recommends default denial and permission validation on every request: https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html. Supabase RLS reference: https://supabase.com/docs/guides/database/postgres/row-level-security.

Proposed repository structure

```
fire-and-flames/
  apps/web/src/
    routes/public/   routes/student/   routes/staff/
    features/auth/   features/events/  features/teams/
    features/payments/ features/passes/ features/scanning/
    features/judging/ features/notifications/ features/admin/
    components/ui/   styles/           assets/
  packages/domain/  packages/validation/  packages/contracts/
  backend/functions/  backend/migrations/  backend/policies/
  tests/unit/  tests/integration/  tests/access/  tests/e2e/
  docs/  scripts/  public/brand/
```

Keep shared types, validation rules and state-transition definitions in domain packages. UI calls typed commands; it does not write arbitrary payment/status columns. New project setup is a future development phase, not part of this planning delivery.

Entity model

Entity	Core fields	Integrity rule
profiles	auth user ID, name, email, phone, college, course/year	One profile per identity; own updates only
organizer_units	ID, label, contact	ED Cell and Srustiprosar separate scopes
role_assignments	user, role, organizer/event, active window, issuer	No self-assigned privileged role
events	organizer, slug, status, capacity, team limits	Unknown capacity/fee remains null
event_versions	rules, price options, rubric, publishing approvals	Accepted version immutable for a registration

Entity	Core fields	Integrity rule
sessions	event, venue, start/end, reporting time	Explicit timezone; detect overlaps
teams and members	event, captain, membership status, consent	Unique active event membership per participant
registrations	owner, event version, mode, roster snapshot, state	State transition validated; one active entry per participant/event
reservations	registration, capacity units, expires, review hold	Atomically acquire capacity
orders	registration, total_paise, currency, payment profile version	Server-calculated price, immutable after submission
payment_evidence	order, reference, private object, submitted time	Append versions; limited file access
bank_credits	receiving account, settled transaction ID, amount	Unique credit identity; no double allocation
payment_reviews	evidence, reviewer, outcome, reason	No self-review; audit every decision
refunds	order, amount, approval, bank reference, state	Sum cannot exceed verified refundable amount
passes	registration/member, token hash, version, expiry, revoked	Only confirmed eligible membership
attendance	member, event session, action, station, actor	Unique action per member/session
duty_assignments	staff, event, station, start/end	Scan only while authorized and active
checkpoints	event, sequence, instructions	Ordered team completion constraints
checkpoint_records	team, checkpoint, time, marshal, penalty	Unique completion; corrections audited
rubric_scores	judge, candidate, rubric version, marks, lock	Unique judge/candidate/version; bounded marks
result_revisions	event, totals, tie decisions, approval, publication	No overwrite of published history
outbox and deliveries	event key, recipient, channel, attempts, status	Unique notification deduplication key
audit_log and incidents	actor, scope, action, time, reason	Append-only for application roles

Relationships: organizer has many events; event has versions, sessions and registrations; registration has a roster, one active order and many evidence attempts; verified order plus valid roster yields member passes; pass usage creates attendance. Scores belong to a candidate registration and rubric version. Do not put raw tokens or member details inside public URLs.

Command contracts

Command	Request essentials	Server validations	Result
POST /registrations	event ID, mode, member IDs, rules version	Ownership, consent, eligibility, duplicates	Draft/reference or typed conflict
POST /registrations/:id/reserve	expected version, idempotency key	Roster, published price, capacity, schedule	Order and reservation deadline
POST /orders/:id/evidence	transaction ref, approved upload ID	Owner, file validation, state, deadline	Review reference; never paid=true

Command	Request essentials	Server validations	Result
POST /payments/:id/review	outcome, credited transaction, reason	Finance scope, MFA, no self-review, uniqueness	Verified or exception state
POST /scan	token, station, session, idempotency key	Duty, scope, pass validity and replay	recorded, duplicate or denied
POST /scores/:id/lock	complete marks, rubric version	Judge assignment, bounds, completeness	Immutable locked score
POST /events/:id/results/publish	revision, reason	Super-admin approval, locked scores	Published revision and notifications
POST /staff/assignments	user, role, scope, expiry	Grant authority, MFA, active issuer	Audited assignment
POST /payment-profiles/:id/approve	expected version, approval	Second authorized actor, verified destination	Activated profile version

Use consistent JSON errors: code, safe message, field errors and correlation ID. Return 401 for unauthenticated, 403 or privacy-preserving 404 for unauthorized records, 409 for state/version conflicts, 422 for invalid inputs and 429 for throttling. Never return database stack traces or inferable private record details.

Transactions and concurrency

Capacity reservation locks the event capacity record and counts holds consistently within a transaction. Confirming payment atomically records credit allocation, payment review, registration transition, pass records, audit and notification outbox. Solo intake may become payment-verified while waiting for allocation; only the allocation command can subsequently confirm it. Attendance inserts use a database uniqueness constraint and idempotency key to tolerate retries and simultaneous scanners.

Use expected version numbers for admin edits, result approval and payment profile changes. Server rechecks permissions and state at commit time. Idempotency keys are scoped to actor plus command and bind to a request payload hash; a reused key with a different payload is rejected. Background work claims jobs atomically and tolerates duplicate delivery.

Storage and environments

Separate development, staging and production identities, storage and databases. Synthetic data only in tests. Store uploaded payment evidence privately, with short-lived authorized download URLs, randomized names and content checks. Public assets contain only approved artwork. Backups and restore permissions belong to named college owners; deployment rollback must account for forward-only schema migrations. Use append-only migrations and release-specific restore instructions, never run destructive reset scripts against production.

Proposed service objectives

Target 99.5% availability over the event period, read response p95 below 800 ms and mutation p95 below 1.5 seconds at approved load, excluding user camera acquisition. Proposed recovery targets are one hour RTO and fifteen minutes RPO only if the purchased service and tested backup method support them. Otherwise document the measured limits and obtain a decision before launch. Alerts cover elevated scan failures, review backlog, failed jobs, capacity anomalies and payment profile changes.

Security privacy and recovery

Security is a property of implemented and tested controls. This plan does not certify an undeveloped application as secure. Threat priorities are unauthorized staff access, cross-organizer record exposure, forged payments, reused QR passes, hostile uploads and unsafe result changes.

Required controls

Threat	Required control	Verification
Student grants themselves admin	Server-owned role records; restricted provisioning	Direct API and metadata tampering denied
ED Cell admin sees esports finances	Organizer and event scope on reads/writes/exports	Cross-scope negative tests
Revoked volunteer continues scanning	Active role and duty checked on every command	Revoke during session, next scan denied
Fake screenshot creates pass	Reconcile actual credit; independent finance role	Forged receipt stays unverified
One transfer pays two orders	Unique settled-credit allocation	Concurrent approval collision tested
Copied or revoked QR	Opaque random tokens, expiry, revocation, server lookup, identity check	Replay and old-pass tests
Hidden malicious upload	Extension plus decoded content validation, size limits, quarantine	Invalid MIME, SVG script and oversized files rejected
Client changes fees or score	Server calculation, versioned rules, bounded marks	Manipulated request rejected
Attacker brute-forces IDs or OTP	Rate limits, cooldown, generic account responses	Automated abuse tests
Admin tampers with published result	Locked source scores, dual-stage publication, append-only revisions	Correction produces new revision and notification

Authentication and application hardening

Managed identity handles password hashing, verification and reset. Student email verification is the proposed baseline, with a college help-desk process for delivery failures; never silently bypass verification. Require MFA for all staff actions; a UI prompt without server enforcement is insufficient. Supabase reference: https://supabase.com/docs/guides/auth/auth-mfa.

Prefer server-mediated sessions using Secure, HttpOnly, SameSite cookies where the chosen architecture supports them. Document the session model in an implementation decision before coding. Add CSRF protection for cookie-authenticated mutations, strict allowed origins, content security policy, clickjacking protection, TLS, safe redirects and output encoding. Avoid arbitrary HTML in event descriptions; sanitize any permitted rich text. Validate every request and use parameterized database operations. No service-role credentials or email-provider secrets in browser bundles.

Payment uploads initially allow JPEG, PNG and PDF up to a proposed 5 MB limit; verify real type, strip unnecessary metadata where feasible, quarantine until validated, disallow executables/HTML/SVG and scan supported content before staff download. Reject files that cannot be safely inspected rather than treating their filename as proof. Evidence

is not publicly addressable. Reference: https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html.

Data minimization and consent

Collect the minimum to identify and contact participants: name, verified email, phone where operationally needed, college and course/year. College-ID number or visual ID inspection must have a stated eligibility purpose; do not collect Aadhaar or bank credentials. Keep player IDs only for esports when approved rules require them. Do not collect face recognition data or continuous location.

Registration terms explain organizer identity, selected event, fees, team-member consent, payment verification, cancellation/refund policy, conduct, contact and data use. Separate optional publicity/photo consent from required registration acceptance. Captains cannot give optional consent for teammates. Public results use team/display names only unless participants have been informed and consent where appropriate.

Proposed retention: payment evidence 90 days after final reconciliation, participant contact records 180 days after event closure, and audit/financial records according to the college's approved obligations and policy. These are proposals, not legal retention claims. The college privacy owner must approve retention, deletion requests, dispute holds and treatment of any under-18 participants before launch. Age eligibility is not stated in the sources; add a gate rather than invent a minimum age or consent rule.

Privileged changes and recovery

Payment destination changes require two different named super admins and recent MFA. Notify both organizer leads and finance on activation. Role grants, bulk exports, refunds and result corrections require reasons and are auditable. Prevent self-approval and changing grants to bypass a review conflict. Staff who also compete may not judge or reconcile their own event entry.

Recovery uses a named college owner, independent identity verification, recorded reason and session revocation. Do not email passwords or share recovery codes through event groups. Lost volunteer device: suspend duty and revoke sessions immediately; reissue access on a verified replacement device. Leaked pass: revoke and issue a new version. Suspected beneficiary compromise: pause checkout immediately, preserve logs, verify account with college finance, notify affected participants using approved communication.

Incident runbook

Detect and record an incident ID, severity and owner. Contain by revoking sessions, pausing affected workflow or disabling a track. Preserve relevant audit/evidence without broad data export. Assess affected records, recover from tested backups if needed, notify authorized college leads, and communicate factually to affected users under college policy. Record resolution and preventative action. A frontend rollback alone does not undo financial or database changes.

Release security gate

No production launch with a known privilege escalation, cross-scope data leak, duplicate-credit allocation, invalid pass acceptance, exposed secret or unrecoverable data loss. Require reviewed policies for every private table/storage bucket, a secret scan, dependency review, role tests and an incident rehearsal. Do not promise penetration-test certification; commission an independent review if required by the college.

Notification design and message catalogue

Use durable in-app messages as the primary record and verified email for transactional alerts. SMS/WhatsApp are optional channels requiring an approved provider, budget, consent and tested delivery; they are not assumed operational. Never call a message delivered merely because an API accepted it.

Delivery rules

Write the event and outbox row in the same database transaction as the business action. Worker sends at least once; a deduplication key of business event, recipient, template version and channel prevents repeated user-facing sends. Save queued, attempted, accepted, delivered where a provider proves it, bounced and failed separately. Track in-app read state separately from email delivery. Do not expose tokens or payment evidence in messages.

Proposed email retry schedule: 1 minute, 5 minutes, 30 minutes and 2 hours for transient failures; permanent failures go directly to the support queue. Honor provider rate limits. Staff dashboard alerts on exhausted attempts; a retried notification must not repeat the underlying payment or registration action. Cancellation suppresses obsolete scheduled reminders. Send one current schedule change summary instead of every intermediate draft.

Trigger matrix

Trigger	Recipients	Channel	Content and action
Account verification	Account owner	Managed auth email	Expiring single-use verification link
Team invitation	Named invitee	In-app and email	Event, captain, price model, accept/decline
Roster changed	Every affected member	In-app and email	Change summary and consent action
Payment details received	Payer/captain	In-app and email	Under review; no pass promise
Evidence correction required	Payer/captain	In-app and email	Safe reason and deadline
Payment verified, allocation pending	Individual applicant	In-app and email	Payment received; team assignment still needed
Registration confirmed	Each member	In-app and email	Event and authenticated pass link
Waitlist offer	Applicant/captain	In-app and email	Expiry and action; do not auto-charge
Schedule/venue changed	Confirmed affected participants and assigned staff	In-app and email	Old/new details, IST, acknowledgement if critical
Event reminder	Confirmed participants	In-app and email	Approved reporting time and required ID
Duty assigned/changed	Volunteer/judge/teacher	In-app and email	Scope, shift, supervisor and acknowledgement
Review backlog or delivery failure	Assigned admin/finance	In-app; email escalation	Queue link, no private evidence attached
Cancellation/refund update	Affected payer and members	In-app and email	Actual status and support route
Results published/corrected	Participants and assigned staff	In-app and email	Published revision link
Role or beneficiary changed	Security owner, finance and impacted staff	In-app and email	Actor, change summary, escalation contact

Example templates

Payment received: “We received payment details for {event_name}, registration {reference}. Status: under review. This is not a payment confirmation. View the status securely in your account.”

Confirmation: “Your {event_name} registration is confirmed. Organizer: {organizer}. Reporting: {published_reporting_time_IST_or_TBA}. Sign in to view your individual entry pass. Share your pass only with event staff.”

Correction: “Action required for {reference}: {safe_reason}. Please open your registration and provide the requested correction by {deadline_IST}. Do not send passwords or banking credentials.”

Schedule change: “{event_name} has moved from {old_time_venue} to {new_time_venue}. All times are IST. Your registration remains {status}. View the latest schedule before travelling.”

Refund update: “Refund for {reference}: {actual_state}. Amount: {amount}. {completed_bank_reference_if_available}. Contact {approved_support_contact} if you need help.”

Duty assignment: “You are assigned to {event}, {station}, {shift_IST}. Review your duties and acknowledge the shift. Your access is limited to this assignment.”

Broadcasts and service targets

Admins draft broadcasts for their organizer/events and preview recipient count. Critical all-festival notices require super-admin approval. Validate recipients, prevent duplicate sends, escape inserted values and log sender/template/audience. Never display student email lists to recipients. Operational notices cannot be switched off in a way that hides material event changes; optional promotional messages have separate consent/preferences.

Proposed processing target: routine in-app messages within 10 seconds and email submission within 2 minutes after committed action, subject to provider availability. These are engineering targets, not guaranteed inbox arrival. Registration launch requires verified sender DNS, SPF/DKIM/DMARC as applicable, reset/verification delivery tests and a monitored reply/support mailbox.

Event programme and operations

The event dates are documented as 13–14 November 2026. The times and venue labels below are a proposed programme for organizer review, not a published timetable. Confirm rooms, match durations, staffing, ceremony availability and participant overlap before release. Store all displayed times in Asia/Kolkata and persist timestamps in UTC.

Proposed two day programme

Date and time IST	Activity	Operational dependency
13 Nov 08:00–09:00	Staff briefing, stalls/demo setup, scanner and network checks	Venue handover, safety inspection, duty roster
13 Nov 09:00–10:00	Participant reporting and identity/help desks	Confirmed passes and payment exception desk
13 Nov 10:00–10:30	Shared opening and organizer introductions	College programme approval
13 Nov 10:30–11:30	e-Trivia	Quiz rules, evaluator and answer capture ready
13 Nov 11:45–13:15	Brand War	Rubric, jury and candidate slots locked
13 Nov 13:15–14:00	Break and transition	No meal entitlement implied
13 Nov 14:00–16:30	IPL Auction	Pool count, player rules and scoring supplement approved
13 Nov 10:30–16:30	Bazar and Exhibition open in parallel	Entry staffing; presentation slots avoid competitors' conflicts
13 Nov 11:00–16:30	Provisional esports qualifier window in separate arena	Srustiprosar confirms game rules, timing and capacity
13 Nov 16:30–17:00	Daily close, reconciliation and next-day notices	Resolve unsettled attendance/payment issues
14 Nov 08:30–09:30	Staff checks and participant reporting	Refresh duty assignments and session-specific attendance
14 Nov 09:30–11:00	Treasure Hunt	Route safety, checkpoint marshals and clue test complete
14 Nov 11:15–13:00	Exhibition demos and Bazar jury visits	Judge conflicts checked; demos scheduled
14 Nov 11:15–14:30	Provisional esports final window	Final match format and reporting times confirmed
14 Nov 13:00–14:00	Break and open stalls	Confirm venue availability
14 Nov 14:00–15:00	Reserve slot, verified poll close and dispute review	Official Instagram voting window separately approved
14 Nov 15:00–16:00	Score reconciliation and result approval	Locked scores, tie decisions and finance/prize check
14 Nov 16:00–17:00	Awards and closing	College approval, winner verification, prize process

Date and time IST	Activity	Operational dependency
14 Nov 17:00–18:00	Venue handback and financial close preparation	Inventory, incident log and outstanding actions

This proposal avoids direct overlap among the four E-Fest games, but Bazar, Exhibition and esports run in parallel. At registration, detect confirmed participant schedule conflicts across both organizers. If schedule is unpublished, show that compatibility is pending. After a schedule change, notify affected participants and route conflicts to coordinators. Do not promise unrestricted participation in every event.

Staffing proposal

Appoint an institution programme lead and two named super admins; one event lead per organizer; at least two distinct authorized finance reviewers to avoid self-review deadlocks; a technical lead and support deputy. Suggested campus duties: two welcome/check-in desks with two volunteers each, one help/payment-exception desk, a station lead for each active venue, dedicated marshals at every treasure checkpoint and a separate esports room lead. Actual counts depend on venue distances and attendance.

External evaluators are required by the sources for IPL, Brand War, Bazar and Exhibition; internal evaluators for trivia and treasure hunt. Assign judges explicitly, declare conflicts, test score entry and provide a documented paper fallback. A teacher can fill these roles only when assigned. Staff duties have location, time window, supervisor and acknowledgement.

Programme preparation

Before registration opens: approve pricing modes, rules, privacy/refund terms, eligibility, capacities and official beneficiary. Before one week to event: verify venue layout, power, internet, registration throughput and accessibility. Before one day to event: freeze roster changes except supervised corrections, publish reporting instructions, prepare exception ledgers and print contact sheets for authorized supervisors.

Every morning: count staffed desks, test a valid/invalid/revoked QR on real phones, verify network and backup connectivity, review outstanding payments, confirm judges and announce changes. At close: reconcile attendance exceptions, outstanding credits, score drafts and unresolved incidents; revoke completed shifts and secure devices.

Payment desk procedure

Verify identity and registration reference, locate the actual payment state, and direct unresolved bank-credit cases to finance. Volunteers cannot approve screenshots. Finance checks authorized bank records, records the decision and confirms the correct event/roster. If no slot remains after a late credit, escalate reallocation/refund. End-of-day report separates billed amount, verified receipts, pending evidence, unmatched credits and actual completed refunds; none are reported as interchangeable revenue figures.

Judge and results procedure

Publish rubric and tie-break version before activity; train judges with test candidates. Judges draft scores, check them, submit and lock. Event admin reviews completeness and anomalies without silently changing judge marks. A correction requires authorized reopening, reason, judge revision and renewed lock. Admin proposes results; super admin publishes after score/tie verification. Show published revision/time. A later correction creates a new revision and notifies affected participants. Record winner identity and prize disbursement separately from the public leaderboard.

For Bazar popularity, save official poll window and verified evidence; no fake vote or web poll stand-in. For Treasure Hunt, verify checkpoint order, correctness, finish times and penalties; discipline selection uses its own rubric. For IPL and esports, do not run scoring without an approved formula.

Disruptions and fallback

Network outage: pause online success confirmations, use numbered paper attendance records under supervisor control, and reconcile against server uniqueness constraints when restored. Power failure: protect devices, use approved backup connectivity, update reporting times and keep crowd flow safe. Missing volunteer: supervisor reassigns duty through an audited command. Safety incident: follow college emergency procedures and local trained responders; the app records incident status but does not replace campus safety authority.

Payment fraud concern: do not accuse a participant based on a screenshot; quarantine the claim for finance review. Data exposure: suspend affected access and invoke the security runbook. Weather or venue closure: programme lead approves rescheduling/cancellation, super admin publishes and notifications go to affected registrants. Refunds follow the approved policy, not ad hoc volunteer promises.

Closure outputs

Produce per-event confirmed registrations, attendance, final results, award records, finance reconciliation, unresolved refund list, incident summary and staff handover. Remove expired permissions, archive the published programme, retain records according to approved policy and schedule deletion of unnecessary evidence. Optional certificates require confirmed participation criteria, approved signatories and a separate issuance register.

Testing delivery and release readiness

The following plan is relative to 27 October 2026, with the documented festival starting 13 November. It is a proposed sequence, not a delivery guarantee. Do not compress finance, access-control or live-device testing to meet an arbitrary visual-demo deadline.

Delivery phases

Phase	Proposed window	Deliverable and exit condition
0 Decisions and review	27–28 Oct	Approve this pack; identify payment, rules, staffing and sender owners
1 Foundation	29–30 Oct	Separate environments; verified identity; tested role/scope policies; catalogue
2 Registration and finance	31 Oct–3 Nov	Team consent, capacities, payment review, exceptions, passes and audit
3 Operations and communications	4–6 Nov	Staff duties, scanning, judging, results and durable notifications
4 Experience and integration	7–8 Nov	Final source assets, mobile layouts, accessibility and end-to-end flows
5 College acceptance	9–10 Nov	Real-device rehearsal, payment/sender tests, load and restore checks
6 Controlled launch	11 Nov target	Signed release checklist; monitored registration; rollback ready
7 Event readiness	12 Nov	Roster review, devices, duties, runbooks and support rehearsal

This leaves a short public intake window if launch waits until 11 November. College must decide whether to open a tested registration core earlier or revise the schedule. Any phased launch must still pass identity, finance, privacy and pass-integrity gates. Freeze nonessential effects or optional features before weakening a required control.

Work packages

P0: authentication, event content, scope permissions, team registration, payment reconciliation, pass issuance, online check-in, core notifications, audit and support. P1 before event where required: judging, result approval, waitlist, programme changes, finance reports, restore rehearsal. P2 only after core stability: cinematic effects, optional certificates, optional meals, advanced analytics and game telemetry. Results may use a documented controlled manual calculation only if college explicitly accepts that changed scope; do not claim full scoring automation then.

Acceptance scenarios

ID	Scenario	Pass condition
T01	Desktop and phone hero	Both organizers are identifiable; each CTA opens its own route
T02	External-college student	Non-Srusti verified email can complete eligible profile and registration
T03	Teacher sign-in selection	No role elevation; unassigned teacher sees request/access status

ID	Scenario	Pass condition
T04	Invalid team size	Backend rejects 1/5 members for a 2–4 event and 4 for Brand War
T05	Invitation replay or impersonation	Expired/reused/wrong-recipient invite cannot add a member
T06	Duplicate event membership	Same student cannot enter two active teams for one event
T07	Team versus solo fee	Valid team pays ₹99 once; solo mode follows approved ₹59 policy
T08	Unknown esports fee	No checkout or ₹0 default; information page remains usable
T09	Final slot race	One valid reservation; other request gets waitlist/conflict
T10	Client fee tampering	Server returns approved fee and rejects unsupported changes
T11	Fake receipt	No confirmed payment/pass without independent credit reconciliation
T12	Duplicate credit	One settled bank transaction cannot fund two orders
T13	Self-review	Staff payer/team member cannot verify their own transaction
T14	Late payment after expiry	Exception queue; no over-capacity auto-confirmation
T15	Rejected evidence correction	Version history retained; safe correction path and notification
T16	Underpayment/overpayment	Explicit finance resolution; no silent amount substitution
T17	Refund retry	One refund action/reference; total refund never exceeds permitted amount
T18	Pass privacy	QR payload contains no personal/payment data; token not logged
T19	Concurrent duplicate scans	Exactly one attendance record; second response is duplicate
T20	Revoked or old pass	Denied immediately after reissue/revocation
T21	Wrong event or off-duty scanner	Denied on server; no attendee record exposure
T22	Two-day sessions	Valid day-two action distinct from day-one attendance
T23	Offline or camera denied	Clear not-recorded state; authorized manual fallback works
T24	Cross-organizer access	ED Cell admin cannot read or mutate Srustiprosar private records
T25	MFA and revoked role	Missing MFA or revoked assignment blocks protected action
T26	Beneficiary change	Two different approved actors required; every change versioned
T27	Malicious file	Forged type/oversize/script payload rejected; private storage stays private
T28	Judge bounds and locks	Invalid score rejected; locked score cannot be silently overwritten
T29	Result correction	New approved revision and affected-user notifications
T30	Notification outage	Outbox retries; no duplicate registration/payment action or noisy repeat
T31	Schedule conflict	Overlap detected; change alerts target affected participants
T32	Accessibility	Keyboard, screen reader, focus, contrast and reduced motion pass review
T33	Performance	Approved device/network/load profile meets documented budgets
T34	Restore and rollback	Staging restore verified with measured RPO/RTO and schema compatibility
T35	Secret exposure	No privileged key in public assets, logs, repository or built bundle
T36	Export access	Logged, scoped and purpose-limited; formula injection neutralized in CSV
T37	Last super admin	Cannot remove sole active authority without approved replacement

ID	Scenario	Pass condition
T38	Stale edits and retries	Version conflict surfaced; idempotency prevents duplicate writes
T39	Registration closure	New orders blocked; existing paid exceptions routed correctly
T40	Source correctness	Fee/prize totals and rubric maxima match approved source/version

Test layers and evidence

Domain tests cover transitions, fee modes, rubric totals and timing. Database integration tests use separate identities and concurrent transactions to exercise RLS, uniqueness and roles. End-to-end tests complete both student and staff journeys using synthetic records. Live-device UAT includes Android Chrome and iPhone Safari, camera permission, low light, poor network, keyboard and screen reader behavior. The college witnesses a test payment under its control and verified email delivery before collecting real registrations.

Store test ID, build commit, environment, tester, timestamp, actual result, evidence link and defect reference. No test in this pack is marked passed; these are future acceptance criteria. Asset/document integrity checks are separate from application testing. Re-test changed or failed flows; do not imply that a static mockup proves a backend works.

Launch checklist

- College confirms organizer names, dates, eligibility, capacities, fee modes, event rules and award names.
- Official payment profile verified by two approvers; refunds and exception ownership approved.
- Named super admins, admins, finance reviewers, judges and volunteers invited and trained.
- Private-data access policies, staff MFA, audit, bank-credit uniqueness and scan replay tests pass.
- Production sender, reset links, support contact, notification monitoring and bounce handling work.
- Approved schedule, venue, maps, rule supplements and public copy are published.
- No blocking security/finance defect; staging UAT signed by college operations and technical leads.
- Backup restoration, incident response, duty reassignment and paper fallback rehearsed.
- Deployment owner, rollback procedure, monitoring window and event-day support coverage assigned.

Ownership and cost planning

College sponsor approves scope and public policies; organizer leads approve event rules; finance owns bank verification/refunds; technical owner owns deployments, backups and incident response; independent reviewer checks access/payment tests. Budget categories are domain/hosting, database/storage and recovery plan, email delivery, optional SMS, event network/devices, support staffing and independent security review. Obtain current vendor quotes before a rupee estimate; none is invented here.

Decisions risks and source register

Facts from the supplied files and the user's decisions are distinguished from recommended defaults. Unresolved items below prevent only their dependent production workflows; they do not prevent preparing the specification and artwork.

Confirmed decisions

- The user requested planning documents and assets before development; this delivery does not deploy an application.
- Participants may come from other colleges, confirmed in this chat.
- ED Cell organizes E-Fest; Srustiprosar organizes esports, per user direction.
- Use source fees where known and brochure artwork/colors; do not invent unknown esports fees.
- Provide student and teacher sign-in, admin/super-admin control, volunteers, QR management and notifications.

Decision register

ID	Decision needed	Proposed handling	Owner and gate
D01	Official QR, beneficiary and UPI details	Keep payment collection disabled until supplied and independently checked	College finance; before checkout
D02	Esports fees, rules, capacity and prizes	Publish info only; Free Fire/BGMI fee null and checkout disabled	Srustiprosar; before esports intake
D03	Individual ₹59 team allocation	Consent-based coordinator allocation with no additional team charge	ED Cell; before individual checkout
D04	Exhibition award labels	Prefer summary labels after written confirmation	ED Cell/jury; before public awards
D05	Timetable, reporting, rooms and esports dates	Review proposed two-day programme and conflicts	Programme lead; before schedule publication
D06	Refund/cancellation policy and settlement SLA	Explicit policy and accountable finance queue	College finance; before payments
D07	Student eligibility, age and ID checks	Open colleges confirmed; remaining eligibility to be defined	College authority; before registration
D08	Staff roster and two approvers	Named, individually verified accounts and MFA	College authority; before staff activation
D09	Sender domain and support contact	Verified email plus in-app; optional SMS deferred	IT/communications; before notifications
D10	Hosting, domain, budget and restore capability	New isolated project; proposed Supabase plus Vercel	IT owner; before provisioning
D11	Pool counts and unspecified team/capacity limits	Do not infer IPL total capacity or stall-member limits	Event leads; before related intake
D12	Judge aggregation, ties and conditional criteria	Publish versioned calculation policy before scoring	Evaluators; before event play
D13	Retention, publicity and under-18 handling	Approve purpose/retention/consent policy	College privacy owner; before launch
D14	Optional meals/certificates	Disabled unless funded and approved	College sponsor; before advertising

ID	Decision needed	Proposed handling	Owner and gate
D15	Official Instagram page and poll window	Record verified link/window and evidence procedure	ED Cell; before Bazar voting
D16	Official logo masters and naming	Use extracted images now; confirm Srustiprosar spelling and high-res marks	Communications; before final branding

Key risks

Unverified QR risks payment misdirection; address through two-person verification and versioned settings. The short calendar risks rushed testing; release core workflows first only after their gates pass. Shared campuses and simultaneous events risk conflicts; use schedule checks and explicit allocation. Weak connectivity risks duplicate or missing attendance; use transactional scans plus controlled fallback. Manual reconciliation risks backlog; assign at least two reviewers and clear escalation. Unclear individual-fee allocation risks collecting money without valid teams; gate it until approved. Low-resolution source logos limit large-format output; preserve fidelity and request masters when needed.

Source register

S01: C:/Fire&Flames/ABOUT.pdf, 26 pages. Page 1 identity/ED Cell; page 4 promotional pools; pages 7–8 IPL; 10–11 trivia; 13–14 treasure; 16–17 Brand War; 19–20 Bazar; 21 esports identity; 22 background-only; 24–25 Exhibition; 26 ₹36,000 summary. Extracted text and page images were reviewed. Embedded artwork is preserved with source-page references in the manifest.

S02: C:/Fire&Flames/E-Fest GAME RULES & RUBRICS-1 (1).docx. Opening date 13–14 November 2026; full rules, fees, judging weights and prize summary. Use headings as locators; Word page numbers are not asserted without a verified render. The source DOCX was read, not edited.

S03: User instructions in this chat on 27 October 2026: platform scope, two organizers, premium appearance, official QR and cross-college participation.

S04: Crossfire local reference, read-only review of C:/CrossFire/README.md, src/pages/LandingPage.tsx, src/index.css and package.json. Reusable ideas: event cards, mobile navigation, reduced-motion handling and scoped staff operations. No assumption that its live deployment, credentials, rules, free entry, meals, dates or school eligibility apply here. This was not a security audit of Crossfire.

Technical references checked during planning

• OWASP Authorization Cheat Sheet: https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html

• OWASP File Upload Cheat Sheet: https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html

• Supabase Row Level Security: https://supabase.com/docs/guides/database/postgres/row-level-security

• Supabase Multi Factor Authentication: https://supabase.com/docs/guides/auth/auth-mfa

• W3C WCAG 2.2: https://www.w3.org/TR/WCAG22/

• W3C target size explanation: https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum

• Web performance thresholds: https://web.dev/articles/defining-core-web-vitals-thresholds

These references support design targets, not a compliance certification. Validate current SDK behavior and service plans when implementation begins. No legal, hosting-cost or banking-integration capability is assumed from a generic reference.

Approval record template

Record decision ID, selected option, approver name/role, date/time, evidence location and affected specification version. College review should sign product scope, event catalogue, operational policies, access matrix, design direction and release gates. Keep subsequent changes in a versioned change log with impact on fees, participants and tests. Never backdate approval or label this draft as college-approved.